SIMATS ENGINEERING
ASSESSMENT TOOL 3: Application Based Questions

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Dave's Taxonomy: Manipulation (Level 2) Total Marks: 25

Question Count: 5

Code of Conduct

I K Nandha kumar _(1921260223)___ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: __K Nandha kumar _

Assessment Tasks

Application Based Questions

1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.

2. An organization wants to store large volumes of backup data securely in the cloud.

3. A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.

4. A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.
5. A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments. Provide an example of a microservices-based application.

Application Based Questions Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Relevance to Application	25%	Response directly addresses the given use case with strong relevance	Mostly relevant response	Partially relevant	Weak relevance
Use of Cloud Concepts	25%	Applies appropriate concepts accurately	Mostly accurate application	Basic application only	Incorrect concept usage
Practical Insight	20%	Shows strong real-world understanding	Shows reasonable practical understanding	Limited practical insight	No practical insight
Completeness	15%	Response covers all required aspects	Covers most aspects	Covers only basic aspects	Incomplete response
Clarity	15%	Clear and concise answer	Generally clear	Some confusion in expression	Poor clarity

CO2 -

ASSESSEMENT TOOL-2

Scenario - Based Questions

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course code : CSA1519

course name : cloud computing and Bigdata Analytics.

1. **Cloud Migration Strategy :**

* Cloud migration is the process of moving applications and databases from an on-premises data center to the cloud.

* The organization should first assess its existing applications, databases and dependics before selecting a suitable cloud platform.

* The six migration are Rehost, Refactor, Replatform, Repurchase, Retive and Retain.

* **Performance & Risk Reduction :**

a. Application performance can be validated using load testing, stress testing and performance monitoring. Regular Backupes, Phased migration, pilot testing, security checks and rollback plans help to reduce risks and downtime.

* **conclusion :**

* A paper cloud migration strategy improves application performance. security, scalability and business continuity while minimizing service interruptions.

* It rollback plans help reduce risk and downtime.

2. **Smartcity IOT Architecture :**

a. A smartcity uses Internet of things (IOT) technology to minotor traffic, air quality, energy consumption and public safety in real time.

*. IoT sensors collect data and sent it to the cloud through IOT gateways, message brokers receive and distribute sensor data, while stream processing analyzes the data in realtime.

* Reduces Hardware purchase and maintance costs.

* Cloud storage Provide secure, reliable and cost-effective backup solutions.

3. CDN and Global Application Deployment :

* A content Delivery Network uses distributed servers to deliver content closer to users.

* Nearby servers reduce the time needed to deliver data.

* Store frequently accessed content near users for faster access.

* Cloud providers use multiple regions and availability zones to improve service reliability.

* CDN's and edge servers provide fast content delivery and a better user experience.

4. Cloud Security and Regulatory compliance in Banking :

* Cloud Providers offer security controls to help banks meet applicable regulations.

* Checks security the activities and verifies compliance.

* Records user activities and financial transactions.

* Detect suspicious activities and possible security threats.

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4. cloud Security and Regulatory compliance in Banking:

*. cloud providers offer security controls to help banks meet applicable regulations.

*. checks security the activities and verifies compliance.

*. Records user activities and financial transactions.

*. Detects suspicious activities and possible security threats.

*. Protects sensitive financial data from unauthorized access.

*. security controls, encryption and auditing help banks to protect customer information and meet regulatory requirements.

5. Containers and Kubernetes:

*. Package applications with their required dependencies and share the host operating system kernel.

*. Run seperate guest OS on virtualized hardware

*. Containers generally start faster and have less overhead than virtual machines.

*. Containers can run across compatible cloud platforms and local servers.

*. Containers improve resource efficiency and the portability, while Kubernetes simplifies application deployment and scaling.

End